

Problem Set 2

Submission:

Wednesday, 25/02/2026, until 13:30, HKT (UTC+08:00), to be uploaded on the Canvas course homepage. Only problems marked with (*) will be graded.

1. The birthday problem

Consider $n \in \mathbb{N}$ people assumed to have the same probability of being born on any day of the calendar year. Assume also for simplicity that this calendar consists of 365 days. We are interested in the event

A_n = “at least 2 people in the group have the same birthday”.

The aim of this exercise is to find bounds for $\mathbf{P}[A_n^c]$ for any fixed $2 \leq n \leq 365 = N$.

- (a) Write down an appropriate probability space $(\Omega, \mathcal{F}, \mathbf{P})$ for this “random experiment”. Represent A_n^c as a subset of Ω .
- (b) Show that for any two given people i and j ($1 \leq i \neq j \leq n$) from the group,

$$\mathbf{P}[i \text{ and } j \text{ have the same birthday}] = \frac{1}{N}.$$

- (c) Using (b), and the known inequality

$$\log(1+x) \leq x, \quad x \in (-1, \infty)$$

show that

$$1 - \frac{n(n-1)}{2N} \leq \mathbf{P}[A_n^c] \leq \exp\left(-\frac{n(n-1)}{2N}\right).$$

Hint: It is helpful to recall that $\sum_{j=0}^{n-1} j = \frac{n(n-1)}{2}$ for $n \in \mathbb{N}$.

- (d) Using the above approximation, find the smallest number $n \in \mathbb{N}$ so that $\mathbf{P}[A_n] \geq \frac{1}{2}$.

2. Bayes' theorem and the law of total probability

A poll about the use of a certain app is conducted online among three different age demographics X , Y and Z , which make up 37%, 52% and 11% of the respondents, respectively. The poll concluded that 50% of group X , 72% of group Y and 89% of group Z use the app regularly.

- (a) Suppose we consider one randomly chosen respondent. What is the probability that the respondent uses the app regularly?
- (b) If a respondent uses the app regularly, what is the probability that this respondent belongs to group X ?
- (c) How many responses do we have to collect until with probability 99.9% at least one person is found that does not use the app regularly? Which suitable assumption do we have to make for this calculation?

3. (*) **Rolling dice a random number of times**

[4 Points]

A fair die is rolled N times, where N is a random number with values in $\mathbb{N} = \{1, 2, \dots\}$. We call A_n the event that $N = n$ and S_M the event that the sum of all numbers shown is equal to $M \in \mathbb{N}$. We will assume the following:

- (1) For every die roll, every number shows up with the same probability (the die is fair),
- (2) For $n \in \mathbb{N}$, $\mathbf{P}[A_n] = \frac{1}{2^n}$.
- (a) Determine an appropriate sample space Ω and a map

$$\mathbf{P} : \mathcal{P}(\Omega) \rightarrow [0, 1],$$

such that $(\Omega, \mathcal{P}(\Omega), \mathbf{P})$ is a probability space and (1) and (2) are fulfilled.

Calculate the probabilities

- (b) $\mathbf{P}[A_2|S_4]$,
- (c) $\mathbf{P}[S_4|A_e]$, where A_e is the event that N is an even number.

4. (*) **Sample size vs. signal strength**

[4 Points]

An urn contains 3 red and 3 blue balls. One of these balls, called **A**, is selected uniformly at random and permanently removed from the urn without the color of this ball being shown to an observer. This observer may now draw successively – uniformly at random and with replacement – a number of individual balls (one at a time) from among the five remaining balls.

Peter may draw a ball from the urn six times, and each time the ball he draws turns out to be red. Paula may draw a ball from the urn 600 times; 303 times she draws turns out to be red, and 297 times she draws a blue ball.

Let R be the event that **A** is red, P_1 be the event that Peter's sequence is realized, and P_2 be Paula's sequence is realized.

- (a) Calculate $\mathbf{P}[R]$, $\mathbf{P}[R|P_1]$, and $\mathbf{P}[R|P_2]$.
- (b) Interpret your result.